Statement

In a grid of square cells, we call the neighbors of a cell the cells directly above, below, to the left, and to the right of this cell. (As the grid is finite, cells on the border have fewer than four neighbors.)

I have a grid of square cells. I will reveal a few facts about it:

1. It has c columns and r rows
2. Each cell contains either a 0 or a 1
3. Each 0 has both a 0 and a 1 as its neighbor
4. Each 1 does **not** have both a 0 and a 1 as its neighbor
5. The first row matches a pattern p and the last row matches a pattern q. Note that this can be the same row, i.e. when $r = 1$. The patterns are strings of exactly c characters, with the following meaning: if the $i-th$ character is 0 or 1, then the i-th cell of the row is a 0 or 1, respectively. Otherwise, the i-th character is * and the corresponding cell can contain anything.
6. My grid contains the fewest number of 1s possible given that the previous facts are true.

How many 1s do I have in my grid?

Input format

The first line of each input file contains the number t of test cases. The descriptions of the individual test cases follow.

Each test case starts with an empty line. The second line contains two positive integers c and r (in this order). The third line contains the pattern p and the fourth line contains the pattern q (each of length c).

Output format

For each test case output a single line with a single number: the smallest number of 1s in a grid that matches all the constraints specified above. If there is no grid satisfying these constraints, output -1 instead.

Subproblem F1 (14 points, public)

Input file: [F1.in](/static/finals2025/F1.11f3e6bff722b6a19c4b319b3ede0210.in) (/static/finals2025/F1.11f3e6bff722b6a19c4b319b3ede0210.in)

$t \leq 10^4$, $c \leq 7$ and the area of the whole grid is small: $r \cdot c \leq 21$.

Subproblem F2 (20 points, public)

Input file: [F2.in](/static/finals2025/F2.a4a75112b4b4c264c6f3d11f6dee4bed.in) (/static/finals2025/F2.a4a75112b4b4c264c6f3d11f6dee4bed.in)

$t \leq 500$, $c \leq 7$ and $r \leq 50$

Subproblem F3 (20 points, public)

Input file: [F3.in](/static/finals2025/F3.e163215dd20a11100b8471f4dba4cd66.in) (/static/finals2025/F3.e163215dd20a11100b8471f4dba4cd66.in)

$t \leq 10^4$, $c \leq 7$ and $r \leq 10^3$

Subproblem F4 (20 points, secret)

Input file: [F4.in](/static/finals2025/F4.b4b7c108941a1f7e4efdcf0633096044.in) (/static/finals2025/F4.b4b7c108941a1f7e4efdcf0633096044.in)

$t \leq 100$, $c \leq 7$ and $r \leq 10^9$

Subproblem F5 (26 points, secret)

Input file: [F5.in](/static/finals2025/F5.98d29c36e69687c6c6ed8a4fba6fab9d.in) (/static/finals2025/F5.98d29c36e69687c6c6ed8a4fba6fab9d.in)

$t \leq 10^4$, $c \leq 7$ and $r \leq 10^{17}$

Example

input

```
3

3 3
*10
1*1

7 5
0**0*01
10*01*1

6 47
*0****
0***00
```

output

```
3
10
69
```

One possible arrangement for $r, c = 3$ is

```
010
000
101
```